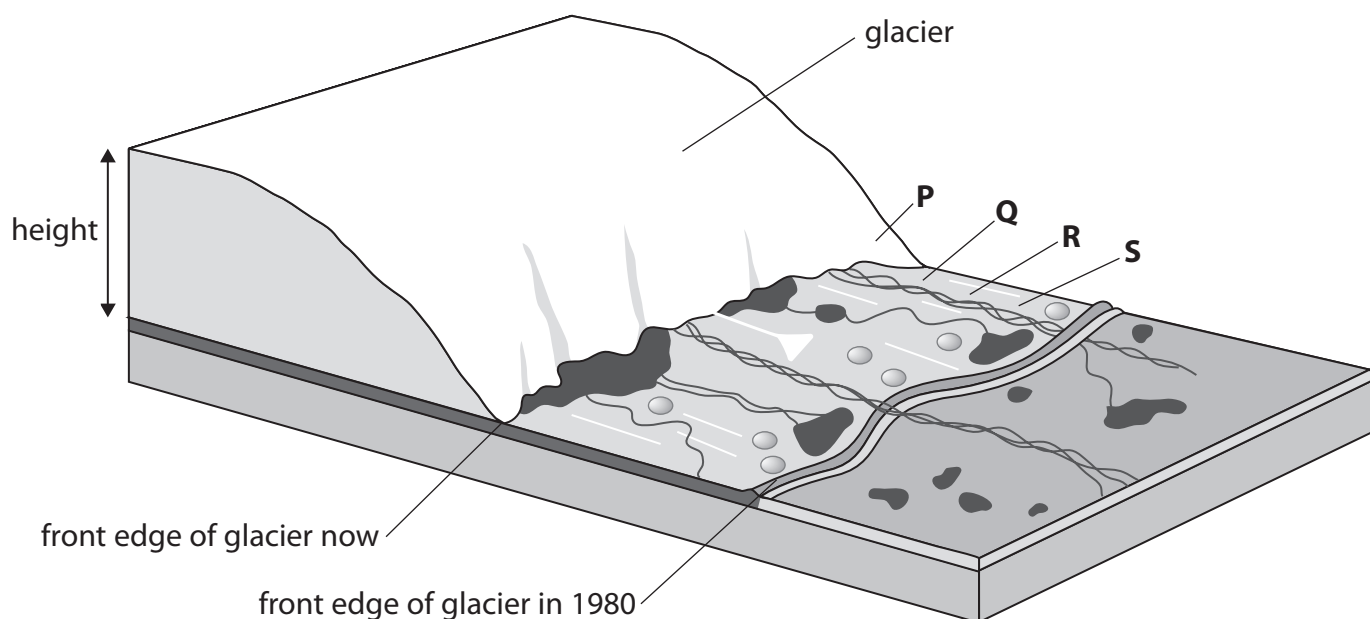
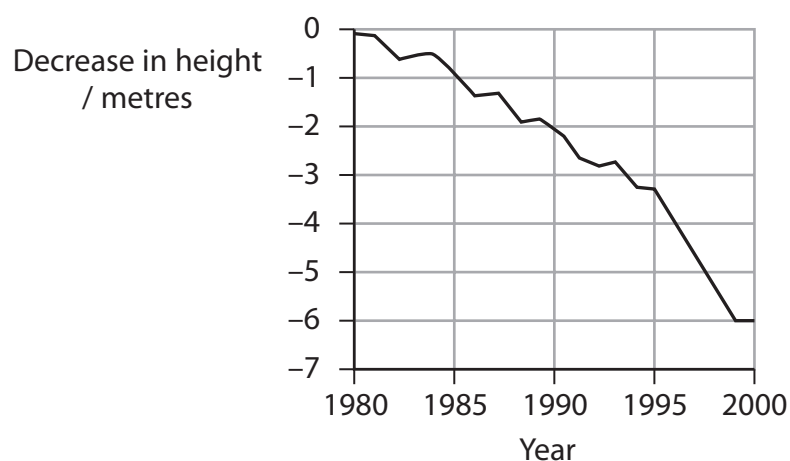


4 Glaciers are decreasing in height and retreating due to global warming.

The diagram shows a retreating glacier.



(a) The graph shows the decrease in height of a glacier from 1980 to 2000.



(i) Calculate the mean drop in height from 1980 to 2000.

Give suitable units for your answer.

(1)

Answer

- (ii) Explain why the decrease in height is greater from 1995 to 2000 than it is from 1980 to 1985.

(3)

- (b) Which letter on the diagram indicates the location of pioneer species only?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

- (c) At different distances from the front edge of the glacier, there will be variations in soil conditions.

These soil conditions include: soil depth, the carbon : nitrogen ratio and the organisms present in the soil.

These differences depend on the length of time the ground has been exposed by the retreating glacier.

- (i) Which term describes the changes in soil conditions and organisms with time?

(1)

- ☐ **A** anthropogenic
- ☐ **B** evolution
- ☐ **C** speciation
- ☐ **D** succession

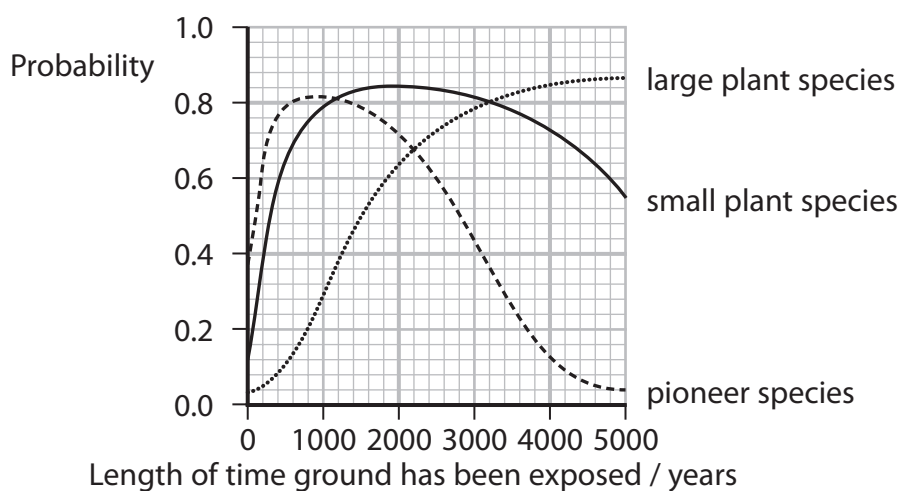
- *(ii) The table shows the soil carbon : nitrogen ratio in ground exposed by a retreating glacier.

Length of time that the ground has been exposed / years	Carbon : nitrogen ratio in the soil
< 200	9.8
2 000	11.6
3 500	10.7
7 200	10.3

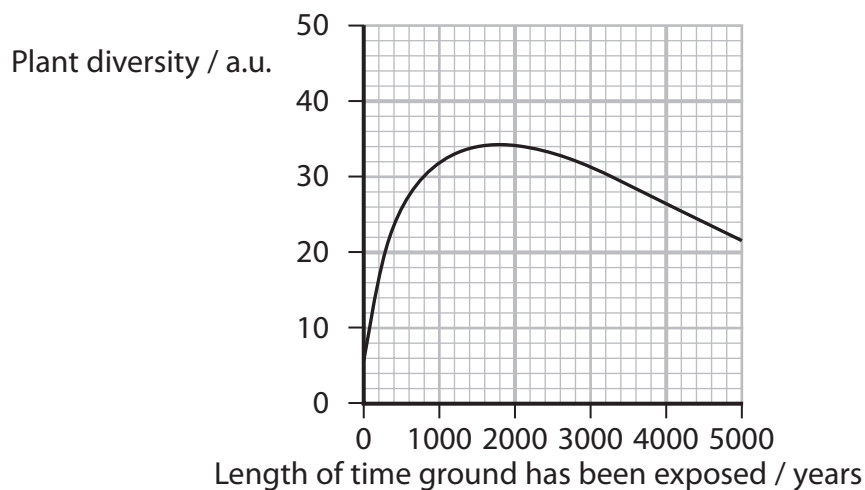
Graph 1 shows the probability that three different types of plant will be found on ground exposed for different lengths of time.

Graph 2 shows a model of plant diversity on ground exposed for different lengths of time.

Graph 1



Graph 2



Explain the variations in soil conditions and the organisms found at different distances from the front edge of the glacier.

Use the information in the question and your own knowledge to support your answer.

(6)

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(Total for Question 4 = 12 marks)